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Profile

Ph.D. candidate in the HMI Lab, School of Computer Science, Peking University, advised by Assistant Professor Shanghang Zhang. My research develops generalizable multimodal agents that perceive, reason, and act across complex physical and virtual environments. I am particularly interested in vision-language-action (VLA) models, world models, and hierarchical architectures for long-horizon decision-making, with applications spanning real-world mobile manipulation, complex multi-stage robotic tasks, and decision-making for game agents. My publications at NeurIPS, RSS, ACM MM, and *Nature Computational Science* have received 300+ Google Scholar citations; I also received a Best Paper Award at the RSS 2026 Embodied4Arts Workshop and have served as a reviewer for NeurIPS 2025, ICML 2025/2026, and ECCV 2024/2026.

Education

Ph.D.	Peking University , School of Computer Science, Computer Applied Technology Research: multimodal perception and adaptive action decision-making for embodied intelligence Focus: Embodied AI, VLA, world models, and VLMs	2022.09 – 2027.06
B.Eng.	Beihang University , School of Electronic Information Engineering Honors: highest undergraduate honor in Beihang's Information Science Division (top 0.3%); highest undergraduate honor in the School of Electronic Information Engineering (top 3%); University Outstanding Graduate (top 10% in major)	2018.09 – 2022.06

Internship Experience

miHoYo , Game AI Research Intern	Beijing & Shanghai, China 2026.07 – Present
- Contributed to a hierarchical high-level/low-level fast-slow architecture: a deliberative module handles semantic understanding, long-horizon planning, and complex decisions, while a reactive module provides low-latency action selection and execution, balancing generalization, reasoning, and runtime responsiveness. - Systematically reviewed external data samples, co-developed multiple versions of data-collection and quality-acceptance standards, and validated the first delivered batch. - Researched structured representations of game scenes and assets and developed a scene representation encoder that converts structured information into multimodal-model inputs, exploring deployment without an additional agent-facing RGB rendering pass. - Built an initial evaluation environment and workflow for assessing model behavior, action execution, and deployment feasibility in representative scenarios.	
Beijing Innovation Center of Humanoid Robotics , Research Intern	Beijing, China 2026.03 – 2026.06
- Designed an embodied memory system to mitigate temporal inconsistency and memory drift in long-horizon video generation. - Built a World Action Model on top of the memory system to improve planning and generation consistency for long-horizon robotic tasks. - Investigated joint world-model pretraining on egocentric human-hand and robot data. - Built separate data-validation and cleaning pipelines for both data sources and integrated them into the organization's existing data-processing workflow.	
Simplexity Robotics (Joint Lab with PKU) , Research Intern	Beijing, China 2025.09 – 2026.02
Optimized whole-body control and mobile-manipulation coupling within a latent-reasoning fast-slow architecture; incorporated joint-torque proprioception and Human-in-the-Loop reinforcement learning for real-world robustness.	

- Contributed substantially to two generations of lightweight egocentric hand-data collection gloves. Both used a ZED 2i primary view, dual wrist-mounted Insta360 GO 3S cameras, and a Unity application for visualization and three-camera timestamp synchronization.
- Gen 1 used magnetic wrist tracking and multiple IMUs for finger capture; Gen 2 replaced magnetic tracking with SLAM and reduced the IMU count. Motion retargeting and accuracy validation preserved finger-joint localization while reducing cost and mitigating wrist-tracker dropouts.
- Incorporated multi-task, multi-view human-hand data into VLA pretraining and downstream training, improving out-of-distribution generalization.
- Deployed Galaxea R1 Lite and Franka Emika Panda robots end to end, spanning real-robot data collection and cleaning, model training, testing, and validation; supported bring-up of the in-house i7 Pro mobile-base and dual-arm system.
- Designed and fabricated custom 3D-printed robot components for rapid structural iteration.

Beijing Academy of Artificial Intelligence (BAAI), Research Intern, Embodied Multimodal LLM Research Center

Beijing, China
2024.08 – 2025.08

- Co-built RoboBrain 1.0/2.0 to improve LLM/VLM spatial understanding and temporal modeling.
- Co-developed **AC-DiT**, an end-to-end VLA method that achieved SOTA performance across multiple mobile-manipulation benchmarks.
- Contributed to **RoboMIND (RSS 2025)**, comprising 107K demonstrations across 4 robot embodiments, 479 tasks, and 96 object classes, plus 5K annotated real-world failure demonstrations and an Isaac Sim digital-twin environment.
- Performed hardware bring-up and real-robot data collection for AgileX Mobile ALOHA platforms and a RealMan RM65-B-V compound lifting mobile manipulator.
- Built datasets and benchmarks across SAPIEN, MuJoCo, CoppeliaSim, and Isaac Sim; developed VR/SpaceMouse teleoperation pipelines and led construction of 100+ objects and 10+ scenes as cross-simulator digital assets.

Research Highlights

AC-DiT: Adaptive Coordination Diffusion Transformer for Mobile Manipulation

Co-developed an end-to-end mobile-manipulation VLA with coarse-to-fine action generation; led core implementation and simulation/real-robot evaluation.

NeurIPS 2025

 OPEN SOURCE ★ 50 STARS

LaST-HD: Learning Latent Physical Reasoning from Scalable Human Data for Robot Manipulation

Aligned unpaired human and robot demonstrations through a shared latent reasoning space, an action-conditioned world model, OOL Glove, and progressive mixed training.

arXiv 2026

Technical Report

HDR: Hierarchical Denoising for Multi-Step Visual Reasoning and Robotic World Action Modeling

Contributed to a coarse-to-fine latent-tree world-action model; improved success over streaming baselines, ran 54.2x faster than bidirectional diffusion, and transferred to real robots.

RSS 2026 Workshop

Best Paper

RoboMIND: Benchmark on Multi-embodiment Intelligence Normative Data for Robot Manipulation

Contributed to a multi-embodiment teleoperation dataset and benchmark with 107K demonstrations, 479 tasks, 96 object classes, and 4 robot embodiments.

RSS 2025

 OPEN SOURCE ★ 2 STARS

LaST-R1: Reinforcing Robotic Manipulation via Adaptive Physical Latent Reasoning

Contributed to LAPO, jointly optimizing latent reasoning and actions; reached 99.9% average LIBERO success and improved up to 22.5% over SFT across four real-robot tasks.

arXiv 2026

Technical Report

 OPEN SOURCE ★ 114 STARS

EchoArena: Learning World Models for Reliable VLA Policy Evaluation

Built an evaluator-centric world model that ranks five VLA policies across ten tasks and three embodiments against real-world outcomes.

CVPR 2026

Workshop

Chain of Thought Prompt Tuning in Vision Language Models

Contributed to a Chain-of-Thought prompt-tuning method for VLMs that improves generalization, transfer, and reasoning across image classification, retrieval, and visual question answering.

arXiv 2023

Group-of-Latents: Perceptual Video Compression at Extreme Bitrates via Masked Latent Generative Modeling

ACM MM 2026

Contributed to causal latent video compression separating intra-frame perceptual anchors from inter-frame synthesis, achieving leading fidelity below 0.005 bpp with temporal consistency.

Implicit Neural Image Field for Biological Microscopy Image Compression

Contributed to an adaptive implicit-neural-representation pipeline supporting arbitrary-shape microscopy images and pixel-level decompression, with controllable compression up to 512x.

Nature
Computational Science

OPEN SOURCE ★ 14 STARS

Open-Source Projects

COMMUNITY CONTRIBUTIONS

CAMEL CONTRIBUTOR

★ 17.6K STARS

Contributed RoboTwin agent tools and zero-/one-shot prompting workflows to the CAMEL multi-agent framework.

Awesome-World-Models MAINTAINER

★ 2.0K STARS

Contributed to maintaining and expanding the community-curated index across major world-model research directions.

Obsidian Image Converter CONTRIBUTOR

★ 821 STARS

Added configurable four-state default image alignment through merged PR #379, released upstream in v1.3.20.

ORIGINAL PROJECTS

Awesome World Models with Memory INDEPENDENT

★ 15 STARS

Created and maintain a 40+ paper, 12+ benchmark taxonomy of memory-augmented world models.

Zotero Obsidian Linker INDEPENDENT

★ 2 STARS

Built a bilingual Zotero plugin that refreshes Obsidian notes while preserving content and bidirectional links.

Skills

- **Agent systems and long-horizon tasks:** VLM/VLA/WAM design-to-deployment; embodied multi-agent tools; fast-slow hierarchies, world models, and memory for planning and low-latency execution in robot/game environments.
- **Robot systems and deployment:** Mobile ALOHA, RM65-B-V, R1 Lite, Franka Panda, and i7 Pro; mobile manipulation and whole-body control, proprioceptive modeling, Human-in-the-Loop RL, hardware bring-up, teleoperation, data pipelines, and end-to-end deployment.
- **Simulation and evaluation:** SAPIEN, MuJoCo, CoppeliaSim, and Isaac Sim; cross-simulator datasets, digital assets, benchmarks, and VR/SpaceMouse teleoperation pipelines.
- **Human data and cross-embodiment transfer:** Co-designed two egocentric-glove generations with magnetic, IMU, and SLAM sensing, synchronized cameras, Unity capture, motion retargeting, and human-to-robot transfer.
- **Engineering and research toolchain:** Python, C/C++, PyTorch, and ROS/ROS2; research-code engineering, reproduction, system integration, and custom 3D-printed robot-component/fixture design and fabrication.

Honors and Awards

University Outstanding Student; Outstanding Communist Youth League Member; Grand Prize Academic Competition Scholarship; First Prize Academic Excellence and Innovation/Entrepreneurship Scholarships; Second Prize Student Work Scholarship.

Additional Information

Patent applications

- Real-time collaborative multi-camera analysis for specified targets.
- Joint analysis using fixed ground cameras and mobile aerial drones.

Software copyrights

- Integrated ground-air video retrieval and trajectory playback.
- Integrated ground-air real-time and offline pedestrian/vehicle tracking.
- Integrated ground-air intelligent surveillance platform.

Student leadership: Class monitor and Communist Youth League secretary at Peking University's National Engineering Research Center of Visual Technology; School of Electronic Information Engineering class monitor and Information Science Division academic affairs representative at Beihang University.

Interests and activities: **Badminton:** Peking University School of Computer Science team; former captain of Beihang University's School of Electronic Information Engineering team. **Photography:** Former student journalist with Beihang University's Party Committee Publicity Department and officer in the Beihang Photography Association.